

ActInf Livestream #014.1 ~ The Math is not the Territory: Navigating the Free Energy Principle

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hello everyone welcome to the active inference live stream it is active inference live stream 14.1 on january 19 2021 welcome to the active inference lab everyone we are an experiment in online team communication learning and practice related to active inference i'm going to share my screen for participants you can find us at our website on twitter gmail youtube or our public key based team or shared username this is a recorded and an archived live stream so please provide us with feedback so that we can improve on our work and whether in the live chat or the comments just feel free to share whatever you're thinking all backgrounds and perspectives are welcome here and as far as video etiquette for live streams we'll mute if there's noise in our background raise our hands so that we can hear from everybody who wants to share and use respectful speech behavior so for the active inference live streams in 2021 we're going to be having these regular tuesday 7 a.m pacific times and then we're also going to be having these special sessions like the model stream and several other kinds of events that we're really excited to be planning and go to this big red link and you'll find a spreadsheet and in that spreadsheet you'll find the details on past and present and upcoming active inference live streams here we are in 14.1 discussing the math is not the territory with author mel andrews next week we're going to be discussing the same paper for a follow-up paper and then we're moving on to a different paper but for this week and for next week we're going to be in this awesome paper and this awesome group of discussions today in active stream 14 we're going to have introductions and warm-ups then in the sections of 14.1 we're going to be going through the paper the math is not the territory navigating the free energy principle by mel andrews in october and we have some slides with aims and claims abstract roadmap etc but this is pretty much going to be whatever people want to discuss today and whatever questions we want to address in next week as well so if you have any questions just save them and submit them however you can and get in touch if you want to join live okay awesome so let's start with the introductions and we'll just give a short introduction or check in and then pass to

somebody who hasn't spoken yet and especially if it's your first time we would love to hear what your background or interest in active inference is so i'm daniel i'm a postdoctoral researcher in california and i will pass first to mel hi there i'm mel andrews and uh i think for better for worse my uh doctoral work is going to end up being largely about the free energy principle and active inference so that's why i'm here cool pass it to somebody who can then introduce them hi there thank you mel um hi my name is marco uh i'm from holland my background's in psychology and concrete neuroscience and german masters also worked a bit on its independence and my interest in active inference is um well a multitude of things i just i just believe that it really has a potential uh as alluded to and written about by male uh in the paper which doesn't today uh in general just a lot of potential to understand a lot of aspects of the world in society that we sovereign haven't been able to really grasp with alternative minds so happy to be here then passed an opacity to yvonne hi my name is juan i'm from moscow russia i'm a researcher in system school i'm i'm an engineer systems engineer and i pass it to hey guys i'm shannon and i'm working in i do brain stimulation and i'm also interested in social cognition and how we interact um like in a musical ensemble and how the free energy principle applies to both scales of those questions and um pass it to alex alex we can't hear you distance x alex um but you don't appear muted let's go to alex oh now we hear you go ahead yeah hi everybody i'm alex and i'm in moscow russia and i'm a researcher in systems management school and trying to find a way to find integration between active inference and system engineering approaches and i bless it to alex okay thanks um thanks alex i'm alex um i'm interested in uh i'm a philosopher i guess uh interested in free energy principle and active inference and sort of fundamental questions about mental representation and lately i've been more interested in applied work and sort of um combining eventually maybe combining these approaches with other approaches to machine learning and so active inferences like a technology but um i will pass it on to um let's see uh dave dave you muted so we have dave and then we have also then the last remaining tim stephen i'm retired from information technology live in the philippines my background is in cybernetics and uh machine translation i'm teaching myself neuropsychanalysis and i was so pleased to come to mel's paper because i've been trying to figure out what is this thing after all it's like a a metaphor among metaphors among metaphors and maybe uh you can tell us toward the end of the presentation whether that has anything to do with what you've what you're saying in your paper and i can't even see who else is not spoken uh philippe philippe there we go hi i'm philip uh i'm studying hero tickle biology in prague and i'm computational neuroscience and i joined because i only read

mel's paper and i found it interesting but i don't know anything about it perfect um let's go to tim i'm tim i'm in toronto canada it's my first time joining in the sessions really eager to learn about this stuff was really fascinating and my background is in science and engineering uh specifically i guess electronic systems and software and uh that kind of thing and i'll pass to stephen go for it stephen or fellow jitzer okay and sarah have you gone hi i'm uh sarah davis i'm um in berlin and um right now i'm a master's student in philosophy of science but before that i was an artist and the through line i guess that's interesting to me is um the relationship between information and materials and media so i'm interested in like reservoir computing and how how and when you can separate those things and you know what is life like all these funny things that are circling around that so active inference is just yet another useful thing that um i want to understand better because it seems to connect to a lot of other disciplines cool it'll be interesting to hear about that soon and then i see blue or yes hi good morning i'm blue knight i'm an independent research consultant based out of new mexico nice is there anyone who has not introduced himself otherwise we're just going to keep rolling all right great so let's go to our warm up questions and i'll just put up the first two which are what is something you're excited about today and what is something that you liked or remembered about the paper and so people can just raise their hands and we'll go in order and until i see a hand i think something i liked or remembered about the paper was that there was no figures so from the point of view of usually reading papers with figures that was very notable but then also that there was clarity and structure despite figures which are often used as a uh not quite an aid to structure a paper but it's how we often think about in empirical research like oh what is figure one gonna show but there was clarity in this article without the usage of figures so marco and then anyone else who wants to continue yeah i fully agree that that is um uniquely clear so um i've been kind of lacking behind on on the literature but for me mel's paper is the best paper so far when it comes to contextualizing french principle and like the title implies really mapping it out right contextualizing in the big complicated epistemic landscapes that this all connects to um so it's a huge props to know for that um i think the only other critical piece i've read is from colombo which is more unwieldy um and i think uh the the kind of importance of mapping really shows in this paper because it it opens up a lot of directions to think about the fdp which stands in stark contrast to a lot of other papers are read which kind of uh take it as authoritative and then just move on instead of uh actually trying to interpret it and so mel does a great job at guiding the reader and understanding the nuances and the questions that would arise on when you see those nuances so excited to hear about people's

thoughts today cool very nice uh tim and then anyone else who raises their hand i thought that i was i i noted to myself just you know as i was reading how clear the writing was i thought the writing was really clear and and expressive and explication really well ordered and um and there was kind of like a i guess uh a humility i remember somewhere in this content i saw the idea of the list of epistemic values i think it was you call them and humility was one in the list and that kind of comes through in that paper the way that uh there's no sort of diving for like a discovery or a conclusion it's really just laying it out and and and giving it to you to to leaving the questions open which was really nice yep and also lots of nice philosophy type things like the idea of defending an idea and so it's not common to see in an empirical paper like we're going to defend this it's it's a lot more like a battlefield of ideas in philosophy and in science it's a little bit of a different structure so sarah and then anyone else to like lather on the praise of like uh to mel's paper but yeah the writing was really good and actually when i first read it um i i feel like i understand a lot more than when i first read it and so now i kind of feel like i'm happy to be here because i need to read a second time i've learned a lot more um but yeah the x the exploration of kind of the beginning of the exploration of all these different models it was um uh saw a lot of potential cool and that's kind of part of the idea with a dot one and a dot two is sometimes you read the paper and you don't talk to someone about it you have thoughts one two three and then you have a first conversation about it or you listen to one video and all of a sudden you have thoughts one through ten and then you have the discussion and all of a sudden you're you know up to 20. so that's part of the idea of taking the time to really read through these papers um does anyone else want to add anything i'll throw up the third warm-up question which is what is something you're wondering about or would like to have resolved by the end of today's discussion so alex kiefer then anyone else yeah so i think having this resolve by the end of today is probably a tall order but i'm i'm interested in the question that sort of um explicitly raised it toward the end of the paper about whether the fep is a very abstract sort of uh model of or in some sense maybe representation of nature or a pure formalism and like that's an interesting line so i think it's an interesting question great question sarah then marco yeah i was i to me it all just seems like mel fix it by the end of the hour but um but i yeah i was curious to know like if if you have any insight into ways that the model could never be applied you know like just trying to situate the model in all these worlds that it has potential for cool any other intro thoughts there uh marco and then or anyone else yeah um a lot of questions for mel i think but but i think what maybe i hope so uh possible to to resolve by the end of this stream is um there's only a very brief note about it but but you

also said that it's also compatible with processionism something about possessions i mean they i didn't see anything else about it so i'm not a philosopher so natural kinds for me always was a bit a weird iffy concept for me um but i was wondering if you could resolve maybe um the problems of mapping it to natural kinds of stuff but for professionalism because it seems that activation is much more sexualist kind of framework not really compatible with the essentials objects properties framework of ontology um so yeah i'll leave it at that cool let's definitely return to that and define it and get clarity on it shannon than anyone else hey yeah so i'm really interested in what role the free energy principle or active inference plays in empirical hypothesis testing um and i think this paper did a really good job of explaining that the mathematical models can be very useful to generate questions to generate hypotheses or just to further investigate some empirical phenomenon and i think it would be good this week and next week to just sort of talk about what it means when you say i have a free energy principle characterization of system very nice agreed and um yes we have several fun hours of discussion ahead so i'm just going to flip to this summary slide and ask mel when you're conveying this work to different audiences how do you communicate it or what would you like to start with just how do you give your 30 000 foot overview perhaps to different audiences or perspective yourself in the past yourself in the future however you want to do it yeah i guess i saw a lot of um confusion about what the fep was and what it was doing um and it was my before i could really uh play with it i had to um i had to get on what it was and and make everyone else clear on what it was um and and so i i tried to tackle kind of like the most fundamental questions in this paper about what it is that the ftp is and what it's doing um and in some ways in some ways i think a lot of those remain kind of unanswered by this paper but but no longer unexplored like i i i put it all out on paper so you know so we can talk about it now whereas before there are these things you know like is the fep science science like is this a scientific model or scientific theory or is it you know that the demarcus like does it does it not belong to the demand of science um that that was a big one and no one had asked that um and then uh well alex's question that's what you know and and i didn't come i i i sort of write the first sentence right is saying i think it's the main explains but then i'm gonna i'm i'm i'm writing a paper with an as a photo where we're gonna come down kind of differently on that on the i mean you know it's not written it's not 100 written yet we're not true but but we come down a little differently on the demonstration problem and then um uh in terms of alex's question that was that was one of my main driving questions the notion of as if is the fpp like a really abstract representation of natural systems central processes or is it not a representation

of anything in nature at all and that to me is like totally not answered yet um and there's some there's almost like a queen like that that's almost to me it's like getting to the heart of like naturalism as a as an approach in philosophy like like uh carl's carl friston's line this whole time has been well there's a dissolving line between just math and and physics proper um and that drives me crazy but uh but i'm coming around to a little bit that wasn't a good summary of my paper that was a good summary of of what i'm excited about exploring further fun yep well just a few things in there might fly above or below those who have different um you know listening regimes of attention you said before i could play with it i had to be clear on it and also help other people be clear so that just captured this learn by doing learn by playing and also learn by serving and teaching because those are some of the key pieces and without being playful and understanding that it's a deep well to dive into and then also that there's a community that is also curious those are some of the key aspects and then also it almost goes without saying that this is a really exciting philosophy paper and it's an ongoing philosophical question so people are maybe familiar with a measurement is made on a protein or a new species is found in some forest and it gets nature science and people are really excited about a measurement that's made about nature and then the philosophical questions potentially because they're fundamental or there's no simple sound bite answer or there's no figure there's no graphical abstract you got to read the prose or got to think through it the philosophical questions are often seen as not contemporary or not exciting or not cutting edge but we're seeing in real time with free energy principle and active inference the way that maybe we could even call them top-down philosophical priors on what the fep is are shaping the realization of what emerges from the bottom up for example if it's really from on high and fep is part of science and it's part of a scientific program that's going to be very different several years down the line than if people said this should be thrown over there with the fiction books you know with the biography of somebody else so it's pretty interesting just how these top-down and bottom-up ideas are playing out in the discussion about what this framework is so pretty interesting and anyone can raise their hand or give a thought or a question the other piece i just wrote down until anyone raises their hand was this unanswered but not unexplored which is taking us back to the map metaphor and it's the math is not the territory of course a little bit of a pun on the map is not the territory and what is the map what is the territory and where are their dragons where do we have just the coastline but not know what's on the interior or where are the underlying features of the world stable or changing and then where's our map stable or changing in an attempt to understand and just like the navigation there's

probably pure navigators out there but also there's those who want to move cargo so how are we going to get it done with this map and this territory so we'll go to alex kiefer sarah then mel um yeah so i guess i'm just following up from some of what mel just said um um so so i don't want to take this too far in the direction i'm thinking about because it's it's pretty general so so i to me this this paper raises questions that are uh fundamental like whether or not you're interested in the fep and then it's it's partly a matter of seeing how these two things fit together but like so mel mentioned quine so one problem that i've always had with the with the line that the fep is not falsifiable or that it's like distinct and kind from active inference as a process theory is just like i guess confirmation holism so uh so i just assume that um if this fep thing is informing our empirical theories at all then there must be some semantic logical connections of some kind between the empirical theories and the fep so um and if you're if you're a confirmation holist do you think that you know sensory evidence uh what we observe kind back propagates if you will to like all of the beliefs in our network then i think the fep should be vulnerable to that but again that's not really about the fep so um i just want to sort of raise that issue more explicitly and um i don't know cool well just before we go to sarah could you just define a little bit better what is the confirmation holism oh yeah who who would think differently on this issue and what would be the implications for active inference well so i don't i don't know what people think about this issue these days um actually i'm just this is sort of like a i guess it's attributed to quine and do do him i'm sorry if i'm mispronouncing his name um but the idea is just that you that sort of scientific theory um does it like it meets sensory evidence as a corporate body so like the idea is that if you observe something that's in conflict with your total theory there's no uh i don't know way a priori of telling which of the claims in your theory you need to revise right so like if there's i mean you could you could for example you could say well there must have been an error there must have been a malfunction in the measuring device and so you could you could revise that part of your theory or you could say no this really does falsify my more fundamental assumptions and so that's kind of what i'm getting at and i'm not sure uh if people hold of the alternative view these days that like no there are certain statements that are meaningful but immune to falsification but that's the kind of question i'm raising excellent uh sarah mel marco i think actually just to not overwhelm mel my question could fit in anywhere i'd rather just give mel a chance to yep let's let's go back to mel then sarah yeah the the web of belief thing yeah the the do him applying hypothesis is this idea that that we've got an integrated web of beliefs and you can't test any atomic proposition in isolation from you can't you can't protect anything in your web of beliefs from from

revision with by new evidence which i think it's almost like very uh compatible with with this kind of asian frame view right there you get this you get this spreading its propagation of of evidence through the network all the way up and down um i like that idea but um what did i want to respond to oh yeah someone said someone said active interest is a process theory and i just want to like set the record straight on that um uh the the reviewer one on my magazine i just i just got reviews back on this um it's not published yet i just got reviews back and they pointed out that um it's probably misleading to refer to accident prince of the process theory there have been process theories on the basis of active inference and variational message passing developed so when you see the like the there's a paper fristen schwab uh rivoli joe sulo is on their um 2017 it's called active influence across the street that's that is a process theory that is made from active inference that's that's set in um variational message passing but active inference itself is not a process through so much as the corollary of the fep applies to to to activity in an environment and then processes of that have been made so i i should probably i'm going to go back in and like be more careful with the words i use cool marco then i'll go ahead yeah um i'm not sure if you met me with their process theory but but i i i intentionally also avoided the phrase process theory because i think it's what i was trying to say with that comment earlier is a lot of the treatments of natural kinds and mapping to real states or affairs of seems to me more grounded in a more object property oriented philosophy pathology right but for me it sounds more that's the the ontological uh assumptions or commitments in processionalism are far more compatible with active inference which i wanted to address it later i'm not sure now it's very time but but um uh there's this issue of not demarcation for is fap ai scientific but also demarcation of markov blankets right so as you pointed out rightly there's this dilemma of you know where do you put the threshold for where the boundary of the marked blanket is but for me that fuzziness is not an issue if you would be more professionalist because then the claim with the necessity would be more there is a boundary but that we cannot affirm or confirm where it exactly is what the you know perfect absolute stressful is it's not an issue itself simply that there is heterogeneity there is certain boundarying of states would be sufficient i think because the procession or dynamic nature of these systems uh would mean that you can't say there's a static on simply that there are boundaries between subsystems but but that was not my question my comments um for now was basically one thing i i think is also interesting is um seeing fpai as simply a language so there's the canonical kind of typical notion of math is a language a physics it's a discipline in which that's written in in maths and i i kind feel that this is the same here so daniel gives very beautiful metaphors of

navigation and mapping and to me it seems that uh this whole project and not just theory but the whole project of p and active inference is more of a vehicle of toolkits of languages and allows you to translate back and forth in disciplines translate into mathematical abstraction and as you explore in the mathematical abstract world you can translate back so it's like if you can't see the patterns in empirical data translate back to abstraction find patterns there um see if you can translate them back and then you can test them and so it's like the scientific narrative with a detour but um yeah i don't know if you find it interesting would love to discuss that more later but yeah yeah that it is interesting and uh also the navigation brings us back to cybernetics and the the word cybernetics comes from navigation and so it's almost like if you can't navigate on the water surface and then you go you abstract beyond some sort of limitation and so it's kind of funny because the mapping metaphor again it is a physical one it's related to cybernetics but then also there's this idea of a map in mathematics and a mapping is a function and so there's this abstraction abstraction mapping and then there's also that as a map onto the physical world because we're embodied agents so if mel or anyone else wants to raise their hand otherwise i think there's a few other threads to highlight there go ahead mel yeah i just want to flag something marco said quickly um the markup blanket isn't actually all that it's not actually as a boundary the markup link is not actually all that diffuse and dynamical it's actually it's actually harnessed to like particular particles and you know in a sim so we think when we think about you know the boundaries of an organism or the boundaries of a cell wall or something right you're constantly you are dispersing the cell is dispersing right the material of which the cell the organ the tissue the human being the society is made up right these are these are these are diffuse and they are dissipating they're the material there's a constant turnover in like the material composition of these things right um and and the markov blanket doesn't handle that right now right the markov blanket is ropes to particular particles so to speak in in your system so it's um there there's a there's a kind of a desire in um in the original like burkhoff stuff on dynamical systems um you get you get uh 1927 first off 1927 you get he kind of proposes a notion of wandering sets and um there's kind of a desire carl has a desire to extend the markup blanket so as it so to be able to accommodate where where there is a turn right that where the the material makeup goes away and yet the boundary persists right you get turnover but um it doesn't do that yet so it's actually it's actually it's not nearly as like dynamical as as you'd want it to be i think the the wandering sets and the attracting sets it kind of reminds me again with a navigation metaphor the attracting set could be like there's like a whirlpool and all the boats around it are kind of being drawn into this attracting set if

they move away then they're drawn back in and then the wandering set is like a flotilla of ships maybe some are joining and some are leaving but they're maintaining a coherence as they dynamically are moving through the ocean or as their subunits are changing because there's this dissipation that was brought up and then the living systems that are far from equilibrium are the ones that are apparently successfully resisting that dissolution and so what is it that those living systems are doing and that's the what is life question that's schrodinger 1944 to maxwell 2018 that pipeline so it's it's really interesting questions marco then any other raised hands thanks uh and um thanks for the clarification um maybe i'm misunderstanding something but um so so i guess for me maybe it's my my personal interpretation of it all but for me the primacy is actually less about the formalism as for example the pro and originally uh formulated as but what for me is interesting is more the conceptual interpretation of there's an intermediating kind of interface right there's a unidirectional flow um that's that's in its dynamics allow for this protection of states that shouldn't go beyond certain uh boundary for example um and in that sense it oh wait and one more thing is that maybe that which is to be bounded or protected doesn't have to be a particular state as such maybe it's like a relation for example you're protecting the property of certain nodes of certain states or certain densities that they are adaptively viable for other systems that they might be in relation with um but but i think in general i think what this all shows is that there's a lot of freedom at least in my opinion to interpret the picture that follows from ffp ai i'm still kind of unclear why it's not dynamic because the mathematical nodes can be variably mapped to physical nodes in my opinion because i'm not sure what you mean with particles right so it's something like we we do the boundary centers of particles but more in terms of sensory states or or or that through which um the activation inference is mediated um but yeah that's my but i would love to hear more about you brothers cool if anyone else raises their hand i would just kind of almost take stock there's a few threads happening here there's some philosophy of science questions what is a principle what is a theory what is falsification all these types of questions were um brought up from the more philosophical side of the literature and then there's this a little bit related discussion about markov blankets dynamical systems the definition of empirical measurements and how cleanly we sort them into these different types of uh natural kinds if we want to use the philosophical word or maybe just a functional whole to be less essentialist about it there's a few different threads happening here so it's just good to keep track of these pieces and uh yeah if uh steven i saw your hand raised or anyone else yeah hi thanks um i suppose one thing that i've been quite interested in is dimensionality in the sense that entropy

is a dimensionless sort of property so you get this interesting question that so the free energy principle in a way is mostly dimensionless or it's like some sort of trend and then when that comes into active inference it doesn't get fully dimensionalized well it doesn't it's like lots of different active influence processes each one can contribute to dimensionalities like multi sort of threads the actual signal unlike when you mention like say in cybernetics where you have something that is guiding you the signal itself doesn't have you know as we say the signal doesn't contain per se the data particularly the entropic noise it's through inference of the dynamics within the noise that the generative model can then process that and so it sort of raises this question about you know when things go from one level to another blankets in these states um is that transference between states also dimensionless in that sense or you know where did dimensions start to come in as we go up the blankets or maybe as things get aggregated interesting question and that reminds me of some other questions that have been raised about the dimensionality of active inference like is each sensory modality enshrouded in its own blanket or is it sort of a tuple that gets passed across a single organismal blanket that represents multiple sensory possibilities so interesting questions anyone else want to raise their hand or we can just start to slowly walk through the paper and just any thoughts that people have will raise them so we checked out the abstract in the dot zero video here's the road map so this is a pretty thorough road map because we can see a lot of the sections and a lot of the terms that come into play so mel i have a question what was your intention with writing it you said that you wanted to clarify it and learn as well how did that translate into this structure and then maybe where are you headed you know off this road map this was one journey what is your next journey yeah so a friend brought my attention to um to the free energy principle and i think fall of 2017 which would have been like my penultimate semester in undergrad and i was taking a course on like mathematical psychology so um uh if if you one of my references is right so the reference to this normative model this normative versus descriptive process versus right that that that reference is argon can minus 95 which is um he's he's one of the primary figures in in maths like so a lot of cognitive science you know has had subsumed math like but but the idea of like having these formal mathematical models of um of cognition so that's that's what i was immersed in so so i i was uh that's what i was thinking about most at that time so like oh a new you know a new mathematical model of psychology perfect um and i i i was very skeptical at first and i dove in and i um it it brought me under it was really there's a lot there and um and i really struggled so at first i really wanted to do that too like you know it was immediately apparent to me that there's this big complicated thing that that very few

people understand and there's a lot of widespread misunderstanding of what it is and what it's doing so i tried to do that at first and i came up with like a little uh a little like dossier on on like the basics of the ftp in like some of its early formulations as an undergrad um and i put that up and but it it didn't solve kind of a lot of the bigger uh i guess like philosophy of science questions about what it is and what it's doing um so i this is like a second stab at that and and from here um i i so i was thinking of the fpp one one way it's helped me to think about it is as a model and i don't i i guess i don't view um models as strict natural kinds or like strips the scientific kinds i don't i don't think that there's like a a list of necessarily sufficient conditions for being a model or like an essence to being a model and the fep is either definitively a model or not i think i think it's helped me quite a lot to think of the ftp as a model it's it's reduced my prediction error if you will so think of it as a as a very abstract formal generic target list model um but but you know and i think in a lot of ways thinking of it as a modeling language or a modeling framework is also really fruitful i think that's important there's not there's not a philosophy of science literature on modeling frameworks or modeling languages right um but but i think that's it it's probably more of a modeling framework or modeling language than it is a model per se um and and next so i i'm i'm next i'm going to try to work on a paper with um within us kind of exploring what it is to be a principal right like what's the principle because there there isn't a philosophy to run on principles really right what is the principle um that's next i also um there's been a lot of comparison of the fpp to to natural selection to darwin's theory of natural selection and um when you ask people about that and this is this is sort of the the the core like pro fep the core like we're working on the fpp camp um often draws that comparison and i think that this so so what they mean to be doing there is saying well it's not directly falsifiable to alex's point i think i think everything is sort of falsifiable every everything in science is falsifiable in so much as it's if it if it doesn't get traction it falls out of use if it's not useful then we don't use it right and then it it ceases to be kind of part of our collective consciousness as as researchers and i think that if if the fep fails to be useful you know it falls by the wayside that's that's gets confined to dustbins of history um so so there is that in that sense i think there definitely is that kind of like confirmation holism there right um if it fails to produce useful lower order theories than it it it's all that of use um so so i think that that in in trying to draw a comparison between natural selection and free energy principle people want to say well the fep like natural selection is not directly falsifiable um in the way that natural selection isn't directly falsifiable and it's um it's also supremely useful um it has this widespread kind of both empirical use and you know grand explanatory power and i think

that this is sort of ultimately conflating multiple notions of natural segments so multiple kind of discrete formulations of natural selection one of these due to popper and one of one of them sort of due to to darwin and then it gets formulated in like lewinton and got christmas and stuff like that um and one of them that safely do the proper which popular later kind of recounts as a as a mistake it's an error news i'm thinking um so i want to do that and i i also think there's kind of a metamark because i've been working on models and reification in models so so when we what's happening with the ftp we can think of as reification conceptual reflication or pernicious relocation where you get uh confusing the map for the territory basically perfect thanks for that response and also welcome uh scott we will go to alex kiefer then marco then tim okay um so i guess there are a number of things that i want to say um so as far as okay right so i think what mel just brought up um reminded me that there's a there's a um one way of of i guess i've discussed this with mel and others so i'm um apologies if this isn't in the paper i'm not quite sure i need to look at it again but basically one way in which this thing could fail to be falsifiable is if it's not so much a claim as like a an approach or like a a sort of stance that leads you to make certain claims so like um so to put that in a if you wanted to put a negative light on that you could say it's a moving target but i i don't mean to disparage the ftp at all i think i think um you know one thing that's going on here is that carl fristin is a fairly creative scientist obviously he's sort of trying to do something new and so he's you know this this is a framework uh mel mentioned thinking of this as a modeling language um it's a framework that's still under construction to some extent and um uh so you could say it's well it's not falsifiable because it's not a particular you know mathematical description it's a uh generator of descriptions like that so i'll stop there for now yep and just to draw one quick thought there it's the principle as being metaphor uh the metaphor is between the principle and the language so could anyone falsify english could anyone falsify the idea of english grammar or is english a moving target is it something that can't be falsified but you can make claims under the umbrella or using the english grammar approach that the claim could be useful or not or it could be useful for one person or not for another person so once you get into the specific process theories the specific claims then you're talking empirical data or you're talking about the perspective of individual researchers but then when you pull back to the principal level and mel it sounds really interesting about the work on fleshing out what a principle is it's a little bit more like you can't falsify python but there's programs that don't work okay so marco tim and then dave thanks um so first of all apologies in the fans with philosophers because i'm gonna consciously tread into pragmatism um so so again i'm not a philosopher but

but i'm kind of interested in in the angle of pragmatism on these matters so um reiterating again the theme of of seeing the fringe principle and active ingredients as more of a language and a framework um we also i think obtain an interesting maybe mode of falsification that actually mel just gave us which is um if certain approach or framework or a system of doing science um is selected by the niche that is a scientific enterprise then that is maybe an interesting form uh well the opposite of falsification right so if it doesn't get adopted then that maybe is some kind of classification but then an interesting uh implication here is following the popularity and the strange like popularity you must have uh of sap in so many different circles um is i think a sign that that the scientists or the researchers who have utilized it have experienced an embodied signal of usefulness and then following that comes a question of okay then where does it come from right and then for me what's really really interesting is then then the fap is the perfect kind of approach to answering or approaching that question right so so upon adopting or upon being exposed internalizing the ftp and active influence what changes within so now we get also two kinds of approaches to language not just you can communicate to other people but also how do the different models you you you embody communicate with each other so the unifying uh capacity of free energy principle is not just uh in the abstraction in the papers but also for people themselves at least for me and anecdotally also others that is i think one of the more unique aspects of active influence it really evokes the sense of unification it allows you to use a single framework and lens for a variety of phenomena and theories and to me that is maybe the biggest argument for it being scientific for the merits it evokes are genders in scientific enterprises including their constituents such as people and resurgence um so uh yeah so i would love to hear more from you uh especially philosophers on pragmatist perspectives sorry i've been binging a bit of hasso chan recently so maybe that's why i'm asking thank you marco tim then dave than anyone else who raises their hand uh did you say pragmatist just now yes that was pragmatism yeah a good segue to what i wanted to mention i guess because i was when you mentioned rarification that resonated with thoughts as i read the paper about um whitehead's fallacy of misplaced concreteness you know the idea that this this this knot in the processes that exists that's being revealed um is actually a process and the things that we see in the objects of the world are really just um uh sort of a rarification of the processes and you mentioned natural selection that was another thing that came up but not in the way i don't think you were mentioning it now the way i thought of it in the reading the paper was uh basically the objects as we call them are simply uh processes that we can take a perspective on that survived you know they're there because they survived they

stayed they became they sustained themselves as you like to call it you know self-organized and all that right um yeah so anyway interesting thanks yeah thanks for tying that dave and then alex kiefer yeah mel uh you don't seem to give the exact title of the 1995 loose paper what was that i've got his 95 book but he doesn't seem to ha is it just the introduction to that uh um the tarot window collection oh mel you're muted but look it up and then post it in the youtube chat or comments not the jitsie chat because it will make a noise so post it on the youtube and then anybody who wants to post any of these papers they're mentioning really the place to do it is the youtube comments or send it to me and i'll put it in the youtube video description so that everybody can access it alex kiefer then anyone else who raises their hand uh yeah i guess i was just gonna create some more problems for the simple discussion here um and well in that okay so so pragmatism's on the table and ratification um so um i guess for similar quantum type reasons i'm not sure how to think about these things i think what tim brought up about um perception forgive me if i could get it wrong but perception being something like a ratification of an ongoing process i think is a really good point so like i mean there's a sense in which i'm a fan of ratification because it's like yeah maybe that's just what we're doing when we you know we take some we have a language uh maybe it's an internal neural language or something or maybe it's an external one and we and we take it to be about the world and that's our lens on the world so without rarification in some sense i don't know um how we we would have any any kind of perception of things but i'm sure that there's a way to distinguish bad kinds of ratification for good kinds um and i was going to say something else but it's i completely forget so yeah good point it's kind of like stability and plasticity remember those two top level descriptors of the cybernetic paper that we heard from in the last couple of weeks and it's kind of like they're not intrinsically good or bad because you could be too plastic or too stable and succeed or not and then beyond whether something exists or not humans have this kind of second layer where it's like oh yeah that exists but i don't like it so it's perpetuating itself but then we just don't prefer it and so that's just our agent's level perspective and so it's not just enough to be reified there's something else happening there so marco then sarah yeah oh to build on uh alex's great point so one of my favorite notions is reconstantine ladders or in buddhism they also call uh talk about the simile of the rounds um and i think this is really what it's about right so we're kind of like i like i said we're kind of stuck with reification that's just part and parcel how we engage with the world because the alternative was believing in nothing and then if you literally believe nothing then how the hell are you going to engage with the world right so all you can do is have good re as good as

possible reification or at least worse reifications um and again for me this is this perceptually this kind of approach or at least perspective that we are just engaging in finding better reunifications such that that particular reification or such that that particular um embodiment or incorporation of a model leads to better genetic dynamics and then to better active inference um due to use active infrastructure framework here um yeah sorry yeah cool sarah then mel um i don't know i is this possibly gonna be an embarrassing moment for me but i feel like that's my job um i don't really understand like as i was kind of stepping through fep the math of it there's a there's a point where i'm like eh it seems like bayesianism has got some like it doesn't seem like even the foundations of what feps are a lot of things are based on i mean this idea of of belief and priors and and that something you know something is absolutely true i guess i just wonder mel if you you know if you ran up against that or like if there's a deeper background that i'm probably not aware of in philosophy about the limits of bayesianism um or these kinds of models so that's i don't it's really a vague question but it's the best i could do no that's an excellent question oh sorry yeah i'm nice it is it is a great question go ahead yeah it's an excellent question so there's a um yeah i think i think when i was first grappling with veganism there's a way in which um there's sort of like a laplacian demon smuggled in like there's a there's a view from nowhere smuggled in there's a a perfect knower smuggled him and that would be the case for the fep but we've we've we worked out a way for it to be just a cognizer that's like self-supervising right it's it's knowledge is limited it's access to to the true world out there is is intrinsically limited and yet it's able to self-correct and self-supervise and if you want um i think sort of the best recent explanation of like why this is the case and and how it's able to do that is um yuck hoe's paper and i want to say synthesizer and i want to say something self-supervision and the fish i guess i have to i get i guess i have to drop this link as well let me go do cool yeah that's what's so fun is the discussion yeah i would love it thank yeah the discussion we raised so many fun links and so many connections and then we just make sure to post them so that everybody can asynchronously or synchronously be in the game so marco then anyone else oh no raised hand yeah oh go ahead uh oh yeah sorry um what i'm gonna say oh yeah and i really love that but i i i wanna propose saying instead instead of a perfect knower because laplacian indeed knows everything but for me bajanism is really more about if we would have to make an analogy it's a perfect pragmatic learner so in essence what it is the nordic principle kind of by design says this is the perfect way of knowing if you also add on a lot of assumptions for a particular age but but uh yeah i guess a bit half jokingly you could say it's a fristen steaming you know the perfect dramatic learner um yeah i just i just

ran up against this just in my everyday life i contradict myself in so many ways i can't even keep track you know like one part of me thinks this the other part of things that and they're literally and so i yeah i just i just started realizing my my own limitations with respect to knowledge and how knowledge would even be modeled but that's part of the design i think that's something i often like to point is it's not about having no conflict it's more about given that conflicts will arise how to best cope with them so the adaptive engagement with the world is also reflexive so your own struggles your own challenges your own seeming contradictions are also opportunities to grow and that's again only possible with this demon that that is pragmatically learning all the time i'm curious but hopefully that's a bit more optimistic yeah i'm curious about this perfect demon because i would say that a bayesian updating agent could make a update that helps it survive in its niche or not it could over learn or underlearn and just simply fail to exist in the future and so natural selection again is the hand that just sweeps off the table the bayesian learners that aren't existing and i would challenge you know there's many thoughts on bayesianism and a lot of debates but alternative being what frequentism or some other vaguely unspecified mathematical framework that somebody doesn't want to put a name on to and so i think it's an interesting question how does mathematics and how do formalisms come into play with philosophy at all what would any equation have to say about philosophy but then once you're in that world of trying to make models that have some element of formalism whether it's natural language or whether it's another language then i think you start wandering over towards the idea of a learning or updating agent and not needing to take the baggage that it's a perfect learning agent just that it is an adaptive agent of some kind sarah and then anyone else who raises their hand yeah this gets for me though um this gets into a kind of a i don't know the the clever philosophy word but um you know this is also why i asked a question in a prior week about the markov blanket because it's like i i picture like a monster truck rally where two two fep models are battling it out basically in my own brain or whatever and so it's like where is the so so in that paradigm where you have these kinds of objectifications you know of one fep with another fep like where the um then you have questions around like the boundary between the two and there's this kind of a there's a there's a thickness about all of this um kind of gets me in like an infinite regress so that's yeah fun alex keefer than yeah so i mean i think this this issue we're talking about now is really interesting i uh yeah i was wondering recently whether bayesianism is a is a prior and what that does when you start thinking about it i think this is totally relevant to the the theme of overall theme of the paper because well of course it is but i mean it's directly relevant

because um like it we're talking about things that get to be so general theories theories or frameworks or whatever that are so general that they start to raise really weird questions but one thing i was going to say is just that like i think maybe this is implicit or explicit what people said already but the fact that we're dealing with variational inference well for okay first of all bayesianism even if it's bayes theorem itself is supposed to like descend from god or whatever it's like actually building subjectivity into science more right explicitly than the alternatives i think people mention contrasting into frequentism but then we're taking a further step away from objectivity and saying yeah and each each creature sort of learns its own approximation to that um so i think um i don't think you can eliminate uh claims to objectivity entirely um i don't i and so this may be maybe this is the the best we can do um or at least a step in the right direction very interesting point marco then anyone marco then mel yeah i fully agree um so obviously can't be eradicated and i think that's kind of the beauty of how like alex said uh this act of infant stuff is taking the the rejection of projectivity even further it's really approaching or making at least the the building the foundations for an objective science of subjectivity right and and to to add uh to touch back on the uh of jokey idea of the bistonian demon um like alex said right they create their own particular approximations um they have to have first instantiated particular priorities but more importantly you also noted on i'm sorry also noted on infinite regress so so that's another beauty for me um there's no problem of if need regress because traditionally with homunculi and stuff and controllers or agents or will theirs is yeah but what is controlling that and there's this movement towards smaller and smaller but that's not the case with the scenario because the dependency is not like a vehicle dependent on the driver but it's more assistant dependent on each other and i think that's kind of the beauty um where the agents in the perfect demon scenario it would be more like um minimal dependence on the agent's faultiness and more about how well the niche of the context prepares that agent uh for the niches to be encountered in their life uh and for those you know with attention for buddhism it's also very beautiful how it relates to this idea of the mutually arising or co-origination um it it's just in essence like alex said it's about subjectivity the the the co-dependency of how densities or how states or yeah and identities house densities kind of adapt each other anyways awesome awesome mel then tim than anyone else yeah so i sort of wanna i guess i want to push marco maybe to to bring up a question he raised when you were giving us the this sort of like breakdown of this which i i thought when he raised i was like wow that's that's a really difficult question that i can answer so um i think that would push a very interesting discussion if only because it's a question that i don't know how to answer

and it's a question of um it was something about well aren't aren't physical descriptions just kind of formal descriptions at some level anyways so you were you monica you raised the quest do you remember do you know what i'm talking about no well you raised you raised a question about about um about this line that i drew between a mere you know mathematical formal statistical description of something at a a physical description proper and you sort of challenge that that such a line exist which is which is tristan's line but somehow uh i i love carl but but um somehow the way he has always raised that it feels to me like a cop-out it feels to me like some sort of evasion but when you pushed me on that i was like you know yeah you're right maybe there isn't maybe there isn't a hard line between and and this this sees into alex's question as well um [Music] it's it's it's all this it's all continuous this this question of like well you know uh is math natural like is is math an arbitrary formal system um where we've invented all the rules and it it just has nothing to do with you contingent nature or or is it continuous you know is it is it just descriptions of nature at a very abstract level um is there a is there a hard line between information seriously well you know going full change is is there uh is there a disappearing line between um between inference techniques between techniques for bayesian infants um as a you know just as a statistical technique and thermodynamics right you know statistical mechanics uh those are the questions that i'm like oh man i can't i can't that's that seems to me to be the most pressing question behind all of this to me yes math invented or discovered thermodynamics invented or discovered fep invented or discovered inference overall statistics all these kinds of awesome interrelated questions philosophy invented or discovered tim alex kiefer then anyone else i was just uh just kind of the homunculus infinite thing versus the dependent arising sort of co-emergence of the of the i think that contrast is really something neat that i don't know how to explore it but they seem like opposite ends of you know two poles of the same kind of consideration and i feel uh like i'm repping whitehead here today but it also touches on what he talks about when he talks about the bifurcation of nature that he decries right so i mean um uh so when we talk about taking a perspective and saying that object emerges to my perception first of all because it survived as a self-recreating thing over time you know that it uh you know um and and i did too we're we're both in the same we're both the same thing you know what i mean like it's not like i'm apart from the world the reason it's in my world is because i'm in its world and and the nature's not bifurcated into mind and the physical anyway awesome point tim because we often say in active inference that the organism comes to embody the statistical regularities of its ecological niche and so people look out there at the ecology around them and they can say

okay that's natural but then the organism is also natural and wouldn't the generative model of its niche be just like its hand be just another phenotype that's generative and therefore natural as well so alex then blue then marco um i i might just i might actually ask for now i just wanted to drill down more on this idea of like mathematics uh what what you know pure mathematical formalisms and how they might represent but i think there's a lot of interesting stuff on the table right now i'd rather hear about it at the moment perfect blue then marco so maybe it'll i'll wrap it back for alex because i kind of want to just touch on this dependent arising right or like the mutual arising you know the world exists because it exists in my mind um and also the idea of whether or not mathematics is natural right like is is this like something that we invented or something that we discovered but i would like to make the argument that that really like as humans we are also natural right so if it can be discovered in more than one place at more than one time by more than one human as many strong theories in science are they bubble up together i think that there is some kind of natural arising or naturalness to these kinds of studies thanks blue marco then anyone else thanks um so so uh i think i know what you meant uh with my comments uh no i think but but let me allow uh to push back on myself um so so i think if we're we we only just descriptions um and the only kind of extra qualifiers or properties of these descriptions as such that's relevant and grounded is their etiology or their genealogy where do they come from right so if we already talk about descriptions we already have to be talking about systems that can generate descriptions i don't think there's a hard line uh in terms of an essence right that's intrinsic property but i think the hard line is more about their path their genealogy how did that description came come to be um and then i think we go back to uh language again and i think this is thanks to um liam bright shout out to him who introduced me to karna and he currently talked about valid physical languages um and i think that's kind of the beauty that the fep is maybe not perfect for every state of affair or space of affairs um but i do think it's true that that you can transfer a lot of what we now at least see as valid physical descriptions to the language ftp but more importantly to me the beauty is the generative aspect so if you transfer something into um translate something into active language of active inference then you also transfer the generative aspects to so if you transfer them to active influence and then you explore in the space that's generated there and you see a pattern or something then that hypothesis obtained or derived in an abstract space can be also translated back and so it's the validity of that back and forth translational path that i think is more important to the question of is it a physical description right is it is it physically rooted rootable if you have these valid languages i don't

know if it makes sense yeah so small point uh very small points is um i've always also taken issue with carl saying that we embody statistical regularities about the world not that it's not true but i think it's it's uh only a partial view for me what's more relevant is that there are statistical regularities due to the fact that the world is systematic um and these regularities allow this kind of tethering right so it's for me it's more about embodying a bridge with the world and what you then get at least my motivation is then that leaves a more openness to also give room for the unique aspect to humanity as in humans are weird we're like really weird i don't know why we're so weird exactly but we're really and that weirdness has less to do with cisco regularities and more about what is evoked or generated or cultivated upon embodying this statistical regulators and upon being faced with the challenges of dealing with the world um and so i guess we have some physical regulators of the world and kind of weird irregularities of ourselves uh if that makes sense yes yeah we we swim in those regularities of the niche you know if the fish were just the statistical regularities of water there wouldn't be a fish there so it has to be something a little bit bigger now another uh cool thing is we we've been talking about this uh similarity between the physical ecology and then our social ecology which is also natural so we can think of a physical example of stigma g like the ants that are digging out their nest and they're modifying their physical environment which then changes how their physical behavior is implemented but with this conversation and in the literature especially there's a stigmergy at the level of information the informational niche which has different perspectives within it different agents that have different access to different kinds of information and those kinds of communications in the informational niche are natural and the generative models that we have are natural as well so there's just so many fun ways that we can think about all right well now that we're niche constructing in this info niche what kinds of uh things do we want to put in this construction that we're working on together so just like the ant engineer might say well we could kind of have a corollary to this tunnel that branches off this way maybe there's something useful over here oh we falsified that corollary tunnel it's not useful it retracts so in the social world in this informational world how do we take that natural perspective and also make it useful for the colony scott and then anyone else who raised just wanted to say with regard to the fish i think some of you heard the example i read research a number of years ago where they took a semi-rigid piece of plastic and put it in a stream and then they varied and increased the speed of the stream and ultimately the piece of plastic started undulating with the periodicity that resembled the fish and so they what they realize is the fish doesn't swim through water it actually passes through the water so it was really over

deep time fish are formed by the water now again not the entirety of the fish but also formed by other relationships with other organisms and media but is similar to the way that insects bats and birds independently developed wings because they were flying in the medium of air but i think i think mel has a point yep let's go here yep we'll go mel ben shannon then marco i i point there's a it's like a lovely little video of a of a dead like salmon in a stream and it's it's it's it's behaving in this way that really seems very lifelike but sad um and and i i think i i once like posted this with the caption like is this is this morphological computation and that made a lot of people there was there like people were very up in arms over whether or not that was morphology interesting question and synapses and neurons being physical what isn't morphological computation under certain perspectives shannon then marco than anyone else that was a brilliant picture of what i'm thinking in my head right now and um to figure back on what um both mel and scott just said so the dead salmon or the piece of plastic you know is passing through the water in and you can very clearly see how the environment or the medium that it's in is forming its behavior even though it's dead or it's plastic as you get into more weird creatures creatures with culture like humans even fish that interact in groups and communicate with other fish in a cultural way then you end up adding different layers of medium so you don't just have water you have now the other fish are in your medium they're part of this social organism and then with humans now the other humans are in are are making another social medium that you're interacting in so you have more and more in the terminology like nested levels of environments that you're embodying thanks for that point shannon marco then scott than anyone else yeah i'm not sure if it's appreciated but but i could i'll continue a theme of linking active influence to some spirituality slash eastern traditions it's um so so the notion of the dao right which stands for the way i think it's also for me it always evokes his resonance with active inference um because there's this emphasis on on on this the way the tao of how the universe works efficiently that is unnameable right and i think that that is also related to this issue of reification earlier that there modes there's systematicity there's rules to how the world works but fundamentally it's unnameable the moment you name it it's not the real way right and uh this kind of finding the intermediate part the the one of least effort or as often as described no effort away um this this too is is exactly uh resonant with that picture of the life the dead fish in the stream um so yeah i think that's just in general there's this theme that i find also very exciting how much activation actually helped me appreciate a lot of philosophies and tradition not just eastern but a lot of let's say forgotten wisdoms and forgotten insights from not just religion and not just spirituality but also arts right we often see

this false dichotomy arts as something frivolous in science truth but a lot of insights scientific insights were actually prefigured by a lot of artists into conceptual form and so in my experience active influence or this generalist general um lens or perspective allows us to reappreciate that which has become too decoupled from our modern postmodern over rational of analytic and a way of engaging with the world yeah thank you marco it's almost like dare i say it the math is not the territory so we'll go to scott blue sarah and then alex keefer so a couple other just observations along the lines of what um shannon was alluding to i think in focal um philosopher for co not the pendulum he um talked about pastoral control which was the idea that the persona of a person is formed by their external norms and rules and laws and culture that they're exposed to so the same kind of thing like a plastic toy in a metal mold being formed so that same kind of notion and one of the things i think is interesting about active inference for me is that those differentials what is the motivation for closing differentials and the motivation is de-risking in leverage grossly stated i think so the idea is if you have a differential between you and the environment you want to protect yourself from the harm from that differential or exploit opportunities of that differential that differential is either temperature differential in the case of a heat engine or an information differential in the case of a von neumann or shannon extension of the second law of thermodynamics so those same differentials you have in carnot's equation for heat and cold you need the differentials and information to have market exchange and i think to have beijing exchange so what was what's the status of a bayesian process if you have no differentials between the input and the expectation so along those lines what it seems like is we have these identity stacks this is what i think shannon was alluding to not claude shannon but our shannon here um and both perhaps but that the you take these combinations of things that in the environment that make you who you are we're like bauer birds we collect up these externalities and internalize them for our expectations and there's a whole set of them and then we go out and identify those that are anomalous for our expectations so on another call earlier there's a someone who talked about the difference uh narrative and story and how your internal narrative and your story is being told you have to line up well to the extent that there's a differential that motivates activity of interaction and markets have been described as places where you find price discovery and solution discovery at large scales so the beijing process seems like a way of incrementally identifying the externalities that are anomalous for your expectations and then being able to internalize them is kind of that bayesian process i think so anyway it's something part of the reason that is so appealing to me is that i've been arguing for a number of years that markets

and countries and companies are biological because their iterations and artifacts of structures of biological beings which is us they happen to take on an abstraction in an informational embodiment but they represent those same processes as biological systems in terms of those differentials and trying to render externalities innocuous or exploit externality differences anyway there's a lot this is feeling very like a there's a strange attractor here it feels like that's pulling the conversation in the direction of that um generative systems that i define the last point i define life as being auto catalytic entry entropy secreting systems so these auto catalytic they keep going and entropy secreting they get rid of disorder but the problem is that the neighboring system has to absorb that disorder and so how do you manage disorder among systems is again when shannon arshan and not claude shannon was talking about the different things the different characteristics that's the neighbors good neighbors when they bang into each other in terms of their creating anomalies for each other's expectations manage the anomalies instead of just winning over each other so anyway just a couple of things there thank you scott blue sarah alex marco so i'm gonna skip over what scott just said and go back to what marco had had previously said about um the connections right between active inference and eastern religion so i'm a student of buddhism for many years and i've been wondering right do we in fact have free will like if we're considering karma and also considering active inference so this is something that like i i find this parallel like karmically we're connected we've got collective karma with groups with systems with you know the universe right in this in this uh philosophy so and then we have our own karma like our previous actions and and um you know have have collected collectively formed us right and so when we're faced with a choice like like i wonder do we in fact have free will because all we can really do is like if we're just the product of our karma we only have one choice like it's the choice that we're going to make but obviously that's the only choice that we could have made because of all of our previous karma right so so and in the same way in active inference in thinking about active inference as um like a computational uh from a computational aspect if this is really representative of you know the brain and how how we think and how you know we operate and how we make decisions so thinking about active inference there's really only one decision that we could make right like given the inputs the sensory inputs and like we have this generative model like given our model and given the inputs of the system and our selected policy there's really only one choice that can be made like as a result of all this like collective computation going forward anyway it's just something i've been thinking about for a while blue it's it's awesome and i just checked on search engine karma as multi-scale bayesian

prior the title is not taken yet so anybody who wants to write that one go for it sarah alex keeper then marco yeah just more random into the mix i guess but um you know i was i was reading about like early babylonian um i don't even know if it was philosophers but somebody you know said oh those guys those guys weren't scientists and because xyz and um of course i i disagreed uh but i was just thinking about just generally um evolution of of science and math um and there just seems to be this constant like bifurcation bifurcation and kind of contrasting this ever more complex ecosystem to the fact that we're um really diminishing our like we're going through species extinction in the actual material world but in the model world we're going through this this proliferation and that just like totally trips me out like what does it all mean in terms of um connecting those two things i i often think about the informational versus material and mourn the loss of um i'm more in the loss of analog actually like this is a big thing for me and i don't quite know where to go with it but um at some point when things became digitized we really did decouple um behavior from material in a way that i think is really significant but i can't quite get my head around where it matters and where to where to cut out the joint i love that phrase um and how to how to explore that so if anybody has advice about that also um but yeah thanks sarah it's like marshall mcluhan's work and other media theorists on the innovation of the written word as one kind of uncoupling and you're hinting at even beyond the written word there's a digital uncoupling and how that influences us and especially just you captured it so well there's a diminishing biological diversity in some ways as we modify our environments but at the same time at least over historical time periods we're proliferating there's more songs there's more scientific theories there's all these types of things but what happens if we reduce our uncertainty about the biological world till it's just something that can't support that kind of theoretical proliferation alex keeper then marco then scott oh man um really interesting stuff going on here so i mean i originally raised my hand because i wanted to explicitly say that i do appreciate marco's um connecting the stuff to buddhism and things like that um that's something that like i hope to write about once i've established myself as a serious philosopher a science type person enough that then i can say things about that and no one will react badly um because i think it's awesome um but and and this issue you're bringing up now with like we're kind of enriching the model space at the expense of the physical world is really deep and important i don't know what to say about it but it's there um and uh yeah um i that's uh the thing that i was that i wanted to bring up i figured since we're all talking nicely and getting along about this these big you know interesting large-scale themes that maybe um what we should do is instead argue about like

some detailed annoying so um so so i just wanted to push um this idea that so the reason that i that i don't see the fep as innocent of representation is just this might this might be a sort of because of my provincial view of what representation is but i figure if you think of a mathematical model as like a structure right and then we can sure you have different windows in that structure depending on like the the actual mathematical language you use to write symbols down but maybe this presupposes mathematical realism in some sense but like you've got a structure if you've got any system of equations and then on my view that just represents whatever is isomorphic to that structure and it could be like a idealized physical system i don't know you could argue about applications and where applications come in but i guess i don't see like how you can escape representation um so let's see what what that does if i say that yep and the infamous representation wars earlier on active streams marco then scott then anyone else who wants to raise their hand thanks um uh i'll try to address a lot of points and hopefully meaningful um so i didn't catch everything which you said scott but but i fully agree that there's this huge hugely important issue of dealing with externalities right so so i i have some issue with saying that we simply throw disorder out in the world i think that's maybe a consequence of taking the notion of exuding entropy a bit too oversimplified entropy isn't strictly disordered um because as it's been noted it's all about actually almost pretty much yet every information theoretic measure is subjective it needs to be relative to something something often forgotten but re-emphasize invasionism and now also machine learning because they're discovering that it's a huge issue anyways so the point is that that um first of all i want to also let you that amongst research and active influence this is indeed something that's being addressed right there's for me one of the most exciting applications of active is kind of a philosophy of society or a philosophy of how we engage with each other in the world and how i like to say it and bryce huber also took that word i think of co-regulation we regulate each other so i wouldn't say that that we intrinsically exude disorder in the colloquial sense it's more about the basic knowledge of active infants you know you perceive something you do something the question is to what extent is that problematic do the changes that you enact in the world lead to undesirable consequences that's the only question really um but indeed we can't really address questions systematically in my opinion without a framework such as active and until then we will be myopic until then we will take these externalities for granted and have to wait we're at the mercy of our discovery when it becomes too problematic such as what we saw with industrial revolution great great progress in the terms that you care about then way later you find out the consequences the catastrophic consequences of

externalities at the point than natural revolution um ignored um but you know we're working on um and a small side point to add to that um having said that that our exuding of entropy is not problematic in itself it comes also with a re-emphasis on how the entropy we exude are actually really great think of children you know those children are still learning they've taken all this stuff they don't understand and they just express so much chaos but we love it it stimulates us it apparently nurtures us you know people get happy when they see children playing um and partly because i think when people are doing this chaotic what a chaotic weird messy behavior in the real world we're able to kind of understand what's behind that and i think that's quite a beautiful way to look at it because then you can also see that what maybe messy random behavior for some person might perceived as something stimulating nurturing nourishing uh so so there's a nice kind of ecology going on there um and and as for blue about free will so i i hate the notion free will mostly used in philosophy because it seems it almost always seems like free from determinism right and i think that's that's that's already a trap that you cannot fall into instead it's more about autonomy right the behavior that you that you engage in the actions that you that you that you make to what extent are they driven by factors that are your own factors that are more internal rather than external and then of course you would get the problem of oh well we don't have a self nothing is true internal sure but okay then that is simply because that which is internal is also cultivated because of the external but it doesn't take away the fact i think that we can say when there's a generative process to what extent is set due to idiosyncratic factors um which i think is beautiful because a lot of these idiosyncratic factors are cultivated in part their engagement with other people with other cultural artifacts and so our cultivation of culture of narratives of interactions with each other our practices our rituals these are all participating to cultivate this collective kind of autonomy to what extent can we as a collective shape the world niche construct the world in a way that's more adaptive to us as an agent on multiple skills um so so also i actually don't really like the notion of district choices but more fluid navigation imagine if you're conscious your conscious being the space that which is conscious you're more affluently navigating the density and it expresses itself maybe as a choice but i think that too is the real kind of case we talked about earlier um so yeah nothing go too long on that um yeah and i also really like the notion of uh this uncoupling so for me what the danger is kind of uh to use a word used earlier um there's this body catalytic power to some uh narrative to some practices which shifts uh the the fitness of these practices to a more overly pragmatic mode of behavior one other way i want to want to describe this is basically a lot of our digital technologies create a training set that is too

decoupled from the actual test set that we will encounter in the world um what you see in the kind of rise of some people might know the notion of meat space and service space when you live in service space you're being attuned to a world that isn't conforming the meat space and i think exactly that is a danger and i fully agree with that even though i'm much very much ancient denizen i do think this is a problem that hasn't been fully addressed is dangerous and the fact that we can't with all these smart people here that we can't formulate it properly or characterize properly is another indication that this deserves more attention um yeah so thanks thanks for all the thoughts scott and then anyone else who raises so a small fun point and then a big um fun point so the small fun point is marco you're talking about children throwing off wildness and entropy and that's recreational disorder right because you know that there's a thing there it there's two points on that one one hot sauce is another one of those right because it's it's uncomfortable but people like it right and so on that there's actually a physical there was an article in nature this is a small fun point article in nature magazine that in your senses of taste there are five senses right there's actually two channels good and bad the good channel is sweet which is carbohydrates and umami which is proteins the bad channel is bitter and sour which are poisons typically an unripe fruit and so salt doesn't have its own channel salt goes from the good channel until the level of blood salinity and then switches to the bad channel so the reason for mentioning that is in food you use bitter and sour as highlights as recreational disorder right and so they're and they're because they're not indicative usually of consumables in nature so that was just an interesting side light on that it's a similar to this notion in the dentist there's this research now that if you have a very painful dental procedure you should actually finish up the procedure with a little extra pain but not but a soreness type pain not a difficult pain because the mind remembers the last pain experienced so the the article said do you want a little pain with that and the idea was extra irritant and the ada said they can't because of the do no harm they can't administer extra pain as a matter of policy but the idea was to hijack the sensation the memory of the bad experience with the lesser pain which is kind of interesting so that was the small fun points let me get to the bigger point which i think um is uh it goes back to what sarah was alluding to so this is that idea of symbolic and physical beings okay this is something i've been playing with for a little while it takes a short period to describe and so let me try that so what if put put aside all your notions of causation for a minute i'm going to go back to something that seth lloyd raised for me a guy at mit wrote programming the universe and he asserted in his book that all interactions since the big bang have increased exponentially all interactions in the universe okay so two two particles hit each

other than they get four particles it's 16 particles whatever and then that iterates out to us okay so moore's law you have this exponential increase in transistor density on chips that has led the fifth order effect of that to an exponential increase in human interactions okay because the chips are more ubiquitous they can digitize more interactions we're having this discussion now is not possible before okay so the more interactions are happening exponentially let's go with for a second suspend disbelief and just imagine that all interactions are increasing exponentially okay what happened with humans is physical space did not offer enough vectors of engagement to contain interactions increasing exponentially and so we had to go to symbolic space so there were a couple a couple of other i'll get back to that in a second so think of clonal reproduction versus sexual reproduction clonal reproduction you resemble your parents and for niche space and again this is not intelligent design but stay with me for niche space clones are not sufficiently varied so that if the niche space is changing rapidly the clones won't have as much survivability mutation random mutations can introduce that noise that can allow the accommodation of that change in physical space i think what's happening with humans is that we are becoming information beings increasingly and i say that to people like yeah yeah yeah and then i say hey you know that 401k that you have if anybody has any retirement savings the i say to them do you think that's a pile of groceries or or a house waiting for you to retire that is data that is nothing the system goes you got nothing it's totally symbolic so one of the if and to the extent that interactions have and their artifacts are increasing exponentially throughout time throughout deep time it was inevitable that we moved to information space because the the phase space available for solutions to our problems was insufficient in physical space we needed more dimensionality in order to address things and that's what language does for us it opens up dimensions it opens up abstractions it opens up things called war it opens up things called stock markets it opens up things called insurance law governance all these things that allow for us to be human and the thing that separates us from other organisms to some extent is that language is where the mind exists because that's the symbolic element of who we've become as information beings but i think one of the things that's been i've been playing with just try to understand what this might be is what does it mean to be converted increasingly to information beings and not we're not still have a physical presence but this came up first and last point here this came up first in the identity work i did years ago where people are saying oh you can't control me on the internet i said sure you can a government a sovereign can put you to death they have the monopoly legitimized violence or they can silence you which is we've seen happen now and so the

physical body if i die right now while i'm talking which some people might like with this long thing i'm saying here if i died right now immediately my legal agency to do things on the internet would change and it's an entanglement it's almost like a quantum entanglement because it would happen instantaneously i cannot have agency if i'm dead so anything pre-programmed in my phys in my symbolic self could keep running but the physical world then doesn't respect you having a will anymore so there is an entanglement so as long as governments continue to control the physical body and have the monopoly of legitimate violence they can control the digital body because putting you to death de-animates the in and we have uh dave then sarah then stephen and we'll steer it back to the paper and also get towards the last thoughts that each person wants to share so again dave sarah steven and then okay the um when the kids make a lot of chaos uh you look at the the especially one-on-one games that kids do they are largely dominance games one kid is going to tend to monopolize being the the dominant figure the cowboy the sheriff for a while and then switch back now if they hang with that role for more uh more than about 60 percent of the time eventually it gets actually almost sadistic and it becomes too aggressive the little things get out of hand the dominated kid cries runs off the smart mommy just kind of gives a little pat and lets the kids go on their way the overprotective mom though almost takes over the bullying and makes it a terrible thing which means that kids can't go back and achieve restoration so when the when the the child who has been paranoid doesn't have the reassurance that oh well okay the world does eventually become okay and the kid who's been in the sadistic infantile mode doesn't get the reassurance well even if i was evil mommy for a while i'm not evil mommy anymore now i'm good mommy so another reason to let your kids whack on each other and get dirty okay thank you sarah stephen and then oh um scott's comment kind of reminded me of what i what i unconsciously was bridging um with respect to information scott said something i verbatim i don't remember but basically that um something about language and for me you know language uh languages as with ecosystems as with um species extinction like languages are going extinct and so languages are a really interesting bridge to this you know information space um and it seems that that fact of language is going extinct seems to contradict something you said scott but again because this is i feel like mel's show i want mel to like hook this into whatever she's working on if if that makes sense for her steven tim then mel or anyone else stephen muted all right sorry sorry i'm really i'm just trying i was looking for a paper that i was trying to find on something this okay um so what i was gonna say is that there's there's a lot of stuff coming up here absolutely um and one thing i like about this paper it sort of starts from

the free energy principle and sort of really pays attention to that and the active inference and i think that there is something important like because we get into this conversations about information and stuff is it's like the free energy principle there's a collary for active inference of these stages and different scales and then once that scaling is happening there's also a free energy principle that sort of scales up anyway the natural nature of how these these societies try to work with these models and there's a paper that was i think in the physics review of life and i'm trying to find it but it sort of talks about how we naturally try and reduce free energy over time and i think this is particularly what's happened since capitalism in the last 500 years or 200 years particularly we want to reduce or there's a tendency in different ways to try and reduce the amount of free energy out there and we can do it more and more because we can construct and our focus is constructing um these um larger scale um societies social networks ecosystems a super organism as uh mel talks about it and we're kind we're sort of constructing this niche to such an extent that we just take that and these perspectives on things and the rarification is almost um de facto but there's also probably more traditionally in history and it's probably also true if you think like what do you do when you're out in the middle of the ocean you can't control the ocean so now you have to try and adapt whatever the winds are doing based on what you've got so this the organism like as an active inference process to try and minimize free energy and just to stay attuned and alive in a niche um is is something that the sort of very foundational understanding from physics and chemistry still alludes to you know and then we have these higher the this ability to construct through action and um and this is where we can start to feel that these actions that we do like we talk about that we are information but this is this is the danger of the verification of information because we at some extent because we've built on top of things yeah we kind of construct things in our niche which makes it and it is kind of an informational hole in our niche but we're still a that has to um dynamically engage with that niche and that information doesn't travel like unlike in traditional inactivism that information doesn't come in as like that's a tree it comes in again in amongst all this noisy um statistical variations and we infer from that so there's this question about that danger the information starts to get rarified and is and becomes seen as something being transmitted down the channel but that's only something that effectively emerges once some processing is done so i think i think that's that's what sort of came up for me thank you steven so in our last couple of minutes we will go to tim marco and then anyone else so let's just keep it brief and summary and remember that next week in 14.2 we're going to be on the same paper so no need to cram it in in the last minute maybe raise some questions or ask

somebody about their perspective so that we can mull on it in the next week and then come back next week with the links ready and with the ideas ready with the questions ready and read the paper another time so tim marco and then anyone else yeah just uh briefly uh the child's play maybe it's obvious and it's like what everybody already knows about things but uh it struck me when you were talking about it that that's kind of active inference on hyperdrive right because the the child's play is this you know generating these fanciful hypothesis you know like the imagination is just going and then that puts them in a position to enact with interact with and maximize surprisal as they run all over the place and trip over things right so it's like it's learning on hyperdrive and then the only other thing i want to say is wrapping up was as i was going through and reading the paper and all those attempts to answer the question what is the free energy principle in terms of a category there was one kind of startling reductive idea that came to mind it seemed really silly but i'll throw it out there it could it just be like an idealization like frictionless you know because um it's kind of it's compared to the um principle of at least action right but but if you look at the actual free energy principle through the layers of statistical mechanics like all those layers going into that uh kind of translation uh the free energy principle almost just seems like since we can't really see all through all that cloudiness of statistical and probability and such and um it seems like just an idealization a way to say we can do something with this even though it's this distant thing that's gone through this probability machine but we can do things with it because we have this idealization that we call the free energy principle so it's just another another option for your what it is thing maybe it seems silly i don't yep mel then marco yeah so so interestingly um we've had we've had in philosophy of science we've had series of how science works for a long time and we've gone through many kinds of uh popular accounts and mainstream accounts of how science works and um these have traditionally emphasized um series um um and accuracy right you want you want you want a an accurate true representation of of what's going on in nature right um but uh and and false circulation this sort of thing um the the kind of and rogue view right now the the the idea du jour is is really in a lot of ways focusing on models focusing on idealization focusing on pluralism um there are a lot of great books on this um michael weisberg's simulation and similarity fantastic book great introduction to some of the big debates models um steven downes has a new like tiny little just introduction that just gives you a breakdown of all the literature and models and um angela pertochnick in my department has a fantastic book called idealization in the aims of science and then um so what was funny to me is like i wrote this paper i wrote this paper and then the the math is not the

territory and then a few weeks later someone um uh a professor of my apartment clean line department announced that they were running a course for the spring term um on on this new book by uh called when maps become the world which is actually a fantastic book i'm really enjoying this book and it's touching on all of these things that you guys have been talking about like kind of different modes of representing and do these things represents or not um and how the kind of feedback between our maps of and like cultural differences and how we how we represent things and um where the world is going these kinds of kind of big picture questions and it's a really lovely read um the idea with with models right is that what they do is chiefly idealized right um if if you look at like the the the whims facts um the one side account of of what's going on in modeling right the idea is that um it's it's not meant to be a maximally accurate depiction of some natural system or some natural process um it's not it's fidelity to the world that makes the model useful but in fact it's the idealization it's what gets coarse-grained black box away you know it's it's it's it's about finding is but latching on to some kind of key feature of the system of interest and that's always mediated by kind of what our what our questions are as researchers what our interests are in these natural systems and researches so it's kind of value-laden in a way um and it's discarding the rest it's discarding the information that is inessential to the question at hand so so idealization is at the core of what a model is awesome response so any very rapid final words this is the true lightning round marco scott alex kiefer oh dear i'm not good at lightning around um uh well okay very brief so so i agree about the importance of a notion of friction but i do think it's already uh you get it for free my colleagues say you get it for free with active influence i think because you have two situations where i think you will get friction one is pathologically stuck when you're overly rigid because there's no alternatives of sufficient credence the base factor is too high so you can't move out of it right so instead of saying there's a present friction it's more like there's an absence of a lubricant or something or an absence of a passive and the other side if you have too many possibilities your base factor can't get too high because everything's kind of similar so you get stuck in that and i think for me those are two forms of friction slash inertia um i just very briefly want to wrap up that big theme of how we scaffold all these things upon society which is attractive upon extractions and the many many dangers that lie there indeed you would have i think an auto catalytic pitfall of um trying to cope with the complexity of the world but falling into heuristics and when these heuristics are used to minimize your experience consequences of excessive free energy and then adopt that heuristic then it will spread this kind of self-selecting manner as in the people exposed to these notions such as for

example q anonymic conspiracies they heuristically connect everything together and so you seemingly experience coherent as long as you don't stick too far or commit to proper epistemic virtues and and indeed this will become worse and worse unless we have an alternative a better ground at least in my opinion um and i do want to push back a little bit on this uh mostly true claim that the models are just vitalization but i think what's really really interesting is not just models as presented in papers models as defined in maths but take into account the human enterprise that is science because everyone has to adopt this model not just program it didn't some code way more interesting is what happens when you adopt or incorporate or internalize an idealized model and for me that's the expressiveness vacuum upon incorporating it you will de-idealize it you will fill it up you will make it your own particular and that expressiveness my opinion is what makes it interesting and what mel has really laid the groundwork for it might be but yeah i wish we could have talked more about the paper thanks marco again it's dot two it's not about answering every question in the dot one or even the dot two it's about staying excited about including as many people in the conversation hearing from everybody's perspective if you're curious and listening joining the key base actinlab.public is a great way to get engaged with action not just inference so again single kind of closing thoughts but less personal review paper my ted talk but more what are we going to move forward to into 14.2 not that anyone did that scott alex kiefer steven sled the painter kandinsky said that violent societies yield abstract art and one of the things i always wondered is is the reverse also true is abstraction a form of violence and so that's something i i posit to you that 2008 financial break was abstraction of mortgage instruments that caused a of people who were kicked out of their homes the math so the math of the mortgage was not the territory of the lived experience and the need for housing and so that is where the disconnect happened so something to be aware of in terms of abstraction and the possibility of it being a form of violence and the monopoly of legitimate violence being a definition of the sovereign from mills you might think about the monopoly of legitimated abstraction being a form of sovereignty thanks thank scott alex keefer stephen and then sarah um so one thing i'm still interested in um potentially next week is i guess this probably is like a microcosm of some of the stuff we've talked about but um sort of the jaynesian perspective in thermodynamics and in physics and how this relates to all this um because um i think there's potentially something interesting so i one thing i like about this paper i will be brief here one thing i like is that it goes through sort of the history of the formalism of the fep and where it came from and i think in part that's meant to cast doubt on any direct connection between the fep and physics but at

the same time the jnz perspective in physics is something the fep inherits from and it sort of already maybe gets epistemic stuff into physics at a lower low level so i want to talk about the relationship between um entropy and thermodynamic sense and entropy in the shannon sense if we have a chance thank you we're gonna go to mel um with uh first author privilege and then again anyone can drop off if they need to go but let's bring it to a close together otherwise mel stephen sarah yeah actually um i i would really love to talk about that point that you raised um and my other reviewer i know who it is they find it um my other reviewer uh andrew corcoran actually a great guy um raised that there's this new paper um like gothwald and brown published december 3rd 2020 that is in plus computational biology um and it's uh it's it's really getting into changes so it's really getting into free energy it's really getting into the jameson perspective it's like the first it's like the first time anyone has really delved into james and the fep um so i'd love to discuss that because i haven't i haven't read it um yet and i'd love to discuss that thank you now julian james which jeans is that no et james so so max thank you maximum entropy principle i can i can drop this link in the in the comments below and i'd also like to talk about representation like different different approaches representation and how models represent if they do necessarily excellent steven and i'll go back to just my regular camera steven go ahead and then anyone yeah i think the i would like to hear more on that uh james the entropy the maximum entry approach because i think the whole agonic thing is really interesting i think that's this paper like i say is really tapping into that and uh and i suppose one thing i'd like to say is we talk about models is um we're using models to get the dynamics and there's a slight trap because well that makes sense um but you have to take a perspective and verify it and it's like this modern world we just do it but the the key thing that's also present with the whole flows of information is and is that your models can reveal other ways of knowing which are not perspectival and are not themselves so the interesting thing about dropping something out of the model isn't the model is the dynamic shifts that the graphs show when the model's running so that and that can tap into indigenous ways of knowing because the thing with indigenous people i'm saying earlier is that they're trying to attune to the world not say what the world is and model it and that maybe is what we need to do more to actually survive on this planet as it's an ecological collapse because we might need to sense what is it that we need to dynamically embody that isn't entirely more liberal so you know we could talk about that that's enough for me yes sarah did you have something okay marco any last thought i just want to um echo what's been said by mel and alex i also think that one of the most interesting things especially for the next session to talk more about the

relation with thermodynamic and information theoretic especially because there's one sentence in males that i just want to point out so it says the elements of the framework do not map onto any known features of real-world systems at least not with any more granularity or specificity than the constant dynamics of such systems but i found a very short sentence because it seems that the granularity or specificity of their causal dynamics would be sufficient to act as a model right and i think the really really interesting thing that hasn't been exported enough but but mel griff's a good uh starting ground for is asking more about the relation between the topic entropy where i think we should maybe move more or emphasize the the perspective of interpreting entropy as distribution how fast homogeneous how distributed is it as a tendency to kind of flat out to spread out in the thermodynamic sense and then i think with that conceptual metaphor it's not really a metaphor it's just true but with that concept in mind i think it's much easier to have a conversation about the relationship between information theoretic and thermodynamics that being said looking forward to uh thanks so much mel any last thoughts as first and last author otherwise i'll close it out phenomenal thank you this is great yep it's great mel thanks so much for not just the paper but for engaging with us you know asynchronously on twitter and synchronously on video chat all of the participants there was so much we brought up today and it was just awesome to hear what everyone had to share so if you're listening live or in replay you're part of the community of practice and we want to just have you participate however so if you're curious about something post a question if you want to know how you can participate just reach out to us through any number of these mechanisms other than that on january 26th we will be right back here for 14.2 and we'll deal with some of these questions everybody will come with a few new digested thoughts a couple of polished questions and we will go from there so thanks again and we'll see you next week